

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
```

```
library(lubridate)
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
NTL_LTER <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"), stringsAsFactors = TRUE)
#set date columns to date objects
NTL_LTER$sampldate <- mdy(NTL_LTER$sampldate)
```

```
#2
#create mytheme, make text size bigger, axis text black, and legend on the top
mytheme <- theme_classic(base_size = 14)
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
```

```
## List of 2
## $ axis.text      :List of 11
## ..$ family      : NULL
## ..$ face         : NULL
## ..$ colour       : chr "black"
## ..$ size         : NULL
## ..$ hjust        : NULL
## ..$ vjust        : NULL
## ..$ angle        : NULL
## ..$ lineheight   : NULL
## ..$ margin       : NULL
## ..$ debug        : NULL
## ..$ inherit.blank: logi FALSE
## ..- attr(*, "class")= chr [1:2] "element_text" "element"
## $ legend.position: chr "top"
## - attr(*, "class")= chr [1:2] "theme" "gg"
## - attr(*, "complete")= logi FALSE
## - attr(*, "validate")= logi TRUE
```

```
#set mytheme to standard theme
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperatures in July are the same at all depths Ha: Mean lake temperatures in July are not the same at all depths (or temp changes with depth)
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C

- Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

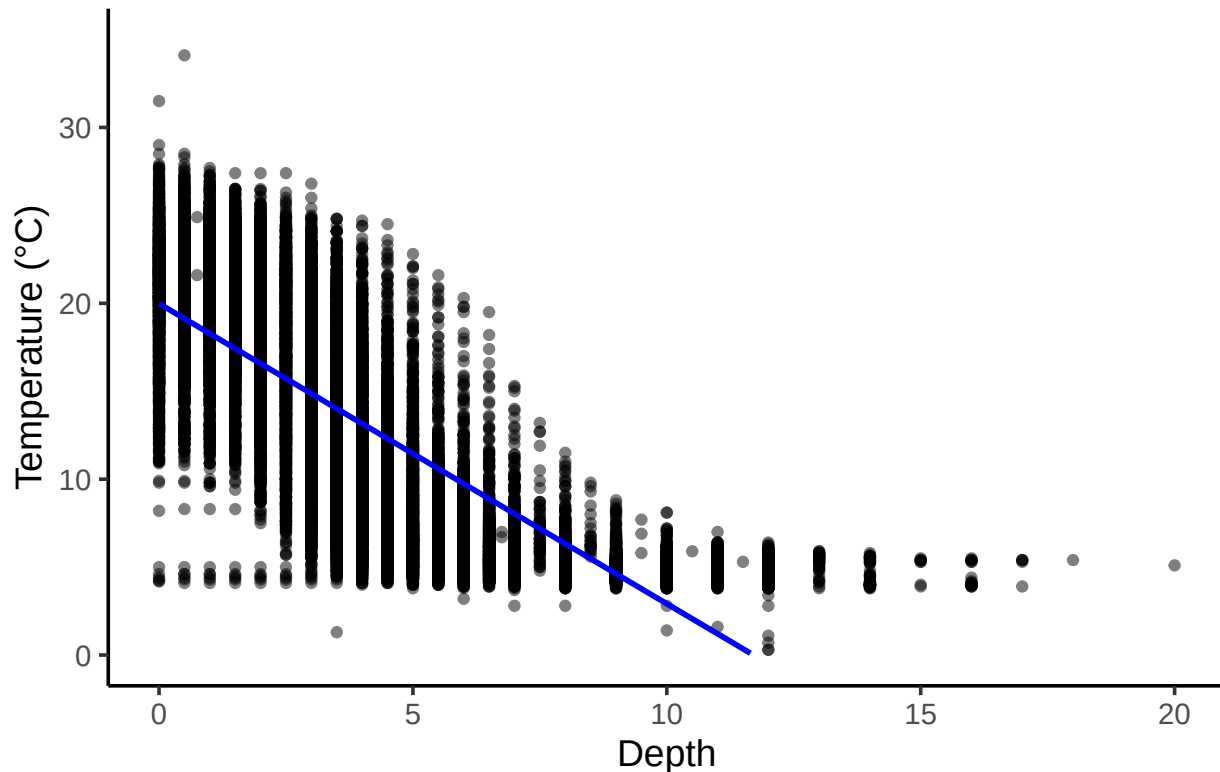
```
#4
#wrapgle data
NTL_LTER_subset <-
  NTL_LTER %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()

#5
NTL_LTER_subset %>%
  ggplot(aes(x = depth,
             y = temperature_C))+
  geom_point(alpha = 0.5)+
  geom_smooth(method = "lm",
             se = FALSE,
             col = "blue")+
  labs(
    x = "Depth",
    y = "Temperature (°C)",
    title = "Temperature change with depth across all lakes in July")+
  ylim(0, 35)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 33 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Temperature change with depth across all lakes in July



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest anything about the linearity of this trend?

Answer: There appears to be a negative relationship between lake temperature and depth, meaning deeper water tends to be cooler. Points at shallower depths (between 0 to 10) are more widely spread, versus at lower depths the points flatten out. This could suggest that the relationship between temperature and depth is not linear.

7. Perform a linear regression to test the relationship and display the results.

```
#7
TempByDepth.regression <- lm(data = NTL_LTER_subset, temperature_C ~ depth)
summary(TempByDepth.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_LTER_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.7864  -3.1363  -0.1219   3.1815  19.2568
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
##
```

```
## (Intercept) 19.986395 0.037166 537.8 <2e-16 ***
## depth -1.707162 0.006366 -268.2 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.961 on 34754 degrees of freedom
## Multiple R-squared: 0.6742, Adjusted R-squared: 0.6742
## F-statistic: 7.192e+04 on 1 and 34754 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The linear regression shows that there is a strong negative relationship between lake temperature and depth across all lakes in July. The p-value is 2.2×10^{-16} , which is less than 0.05 and shows strong statistical significance. The lake temperature is predicted to decrease 1.7°C for every meter of depth. 67.42% of the variance in temperature is explained by the depth, which is based on 34,754 degrees of freedom (the sample size minus 2).

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
temp_aic <- lm(data = NTL_LTER_subset, temperature_C ~ year4 + daynum + depth)
step(temp_aic)

## Start: AIC=92515.66
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 497693 92516
## - year4    1         167 497861 92525
## - daynum   1        47378 545071 95674
## - depth    1       1130140 1627834 133700

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER_subset)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -2.268081    0.007734    0.034990   -1.708651
```

```
#10
temp_best.regression <- lm(data = NTL_LTER_subset, temperature_C ~ year4 +
                           daynum + depth)
summary(temp_best.regression)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.7228  -2.8606  -0.1706   2.9267  17.8338
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -2.2680808   4.5365880   -0.500  0.617111
## year4        0.0077342   0.0022622    3.419  0.000629 ***
## daynum       0.0349904   0.0006083   57.517 < 2e-16 ***
## depth       -1.7086514   0.0060824  -280.915 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.784 on 34752 degrees of freedom
## Multiple R-squared:  0.7026, Adjusted R-squared:  0.7026
## F-statistic: 2.737e+04 on 3 and 34752 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggest keeping all explanatory variables, so year, day and depth. This model explains 70.26% of the temperature variance, which is a slight improvement over the model using only depth that explained 67.42% of the variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
#anova test
tempByLake.anova <- aov(data = NTL_LTER_subset, temperature_C ~ lakename)
summary(tempByLake.anova)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8   57921    7240   155.7 <2e-16 ***
## Residuals  34747 1615571     46
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#lm
tempByLake.regression <- lm(data=NTL_LTER_subset, temperature_C ~ lakename)
summary(tempByLake.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_LTER_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.436  -5.959  -2.559   6.549  24.321
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      16.7363     0.3240  51.660 < 2e-16 ***
## lakenameCrampton Lake    -2.5443     0.3833  -6.638 3.23e-11 ***
## lakenameEast Long Lake   -6.9570     0.3436 -20.248 < 2e-16 ***
## lakenameHummingbird Lake -6.6985     0.4775 -14.030 < 2e-16 ***
## lakenamePaul Lake       -3.9441     0.3316 -11.893 < 2e-16 ***
## lakenamePeter Lake      -4.4838     0.3309 -13.549 < 2e-16 ***
## lakenameTuesday Lake    -6.3896     0.3368 -18.974 < 2e-16 ***
## lakenameWard Lake       -4.3083     0.4395  -9.802 < 2e-16 ***
## lakenameWest Long Lake  -5.6778     0.3423 -16.587 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.819 on 34747 degrees of freedom
## Multiple R-squared:  0.03461,    Adjusted R-squared:  0.03439
## F-statistic: 155.7 on 8 and 34747 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: The ANOVA and lm models return a p-value of 2e-16, so we reject the null hypothesis, and conclude that there is a significant difference in mean temperature among the lakes.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

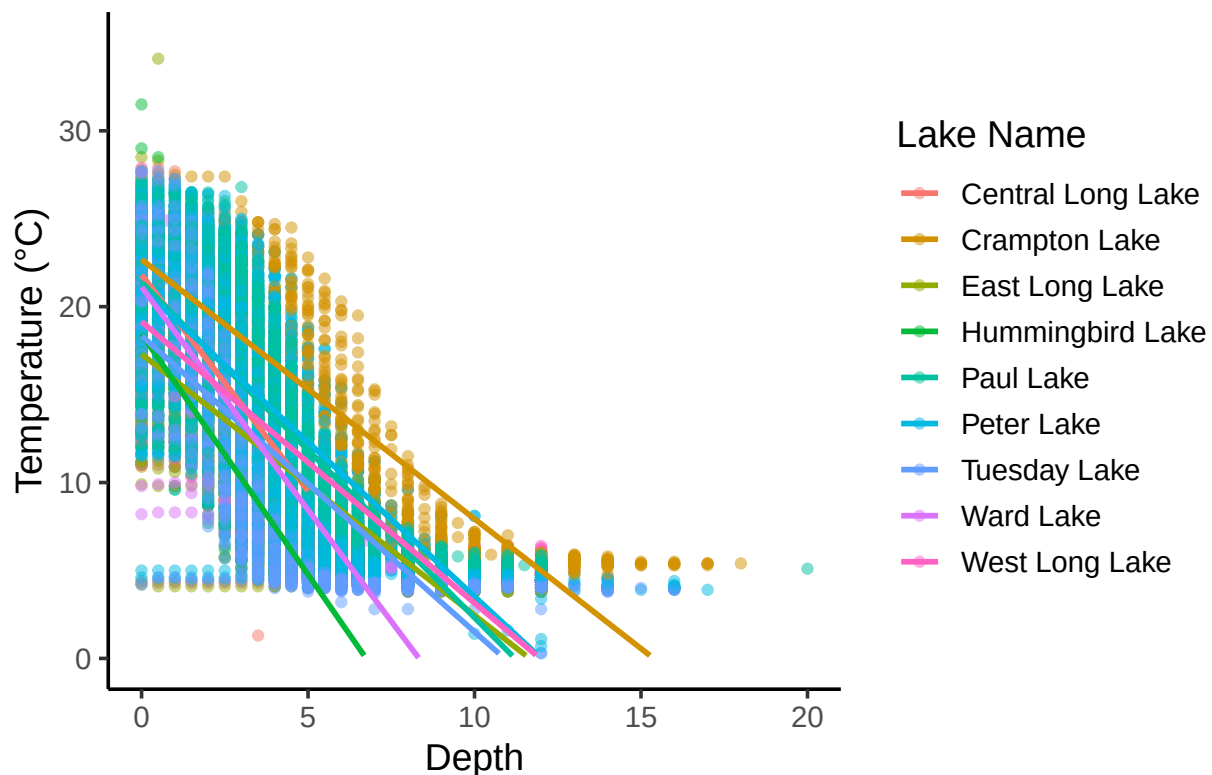
```
#14.
NTL_LTER_subset %>%
  ggplot(aes(x = depth,
             y = temperature_C,
             color = lakename))+
  geom_point(alpha = 0.5)+
  geom_smooth(method = "lm",
             se = FALSE,
             aes(color = lakename))+
  labs(
    x = "Depth",
    y = "Temperature (°C)",
    title = "Temperature change with depth across all lakes in July",
```

```
color = "Lake Name")+
ylim(0, 35)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 126 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Temperature change with depth across all lakes in July



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
```

```
TukeyHSD(tempByLake.anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL_LTER_subset)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.5442854	-3.7331780	-1.3553927	0.0000000
East Long Lake-Central Long Lake	-6.9570473	-8.0227648	-5.8913298	0.0000000
Hummingbird Lake-Central Long Lake	-6.6985124	-8.1794348	-5.2175900	0.0000000

## Paul Lake-Central Long Lake	-3.9440682	-4.9727040	-2.9154324	0.0000000
## Peter Lake-Central Long Lake	-4.4837864	-5.5102613	-3.4573116	0.0000000
## Tuesday Lake-Central Long Lake	-6.3896413	-7.4341675	-5.3451152	0.0000000
## Ward Lake-Central Long Lake	-4.3082596	-5.6715463	-2.9449730	0.0000000
## West Long Lake-Central Long Lake	-5.6777623	-6.7395105	-4.6160141	0.0000000
## East Long Lake-Crampton Lake	-4.4127620	-5.1405823	-3.6849417	0.0000000
## Hummingbird Lake-Crampton Lake	-4.1542271	-5.4140285	-2.8944256	0.0000000
## Paul Lake-Crampton Lake	-1.3997828	-2.0721371	-0.7274286	0.0000000
## Peter Lake-Crampton Lake	-1.9395011	-2.6085446	-1.2704575	0.0000000
## Tuesday Lake-Crampton Lake	-3.8453560	-4.5417779	-3.1489341	0.0000000
## Ward Lake-Crampton Lake	-1.7639743	-2.8831342	-0.6448143	0.0000357
## West Long Lake-Crampton Lake	-3.1334769	-3.8554727	-2.4114812	0.0000000
## Hummingbird Lake-East Long Lake	0.2585349	-0.8857499	1.4028198	0.9987916
## Paul Lake-East Long Lake	3.0129792	2.5954288	3.4305296	0.0000000
## Peter Lake-East Long Lake	2.4732609	2.0610627	2.8854591	0.0000000
## Tuesday Lake-East Long Lake	0.5674060	0.1121132	1.0226989	0.0035472
## Ward Lake-East Long Lake	2.6487877	1.6614645	3.6361109	0.0000000
## West Long Lake-East Long Lake	1.2792850	0.7857610	1.7728091	0.0000000
## Paul Lake-Hummingbird Lake	2.7544443	1.6446129	3.8642756	0.0000000
## Peter Lake-Hummingbird Lake	2.2147260	1.1068972	3.3225548	0.0000000
## Tuesday Lake-Hummingbird Lake	0.3088711	-0.8157039	1.4334461	0.9952041
## Ward Lake-Hummingbird Lake	2.3902528	0.9647057	3.8157999	0.0000071
## West Long Lake-Hummingbird Lake	1.0207501	-0.1198389	2.1613391	0.1224797
## Peter Lake-Paul Lake	-0.5397183	-0.8434372	-0.2359993	0.0000013
## Tuesday Lake-Paul Lake	-2.4455731	-2.8056140	-2.0855323	0.0000000
## Ward Lake-Paul Lake	-0.3641914	-1.3113688	0.5829859	0.9582889
## West Long Lake-Paul Lake	-1.7336941	-2.1410071	-1.3263812	0.0000000
## Tuesday Lake-Peter Lake	-1.9058549	-2.2596746	-1.5520351	0.0000000
## Ward Lake-Peter Lake	0.1755268	-0.7693033	1.1203570	0.9997136
## West Long Lake-Peter Lake	-1.1939759	-1.5958003	-0.7921515	0.0000000
## Ward Lake-Tuesday Lake	2.0813817	1.1169709	3.0457925	0.0000000
## West Long Lake-Tuesday Lake	0.7118790	0.2659563	1.1578017	0.0000259
## West Long Lake-Ward Lake	-1.3695027	-2.3525401	-0.3864652	0.0005266

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Ward Lake statistically has the same mean temperature as Peter Lake with a p-value of 0.9997. Central Lake and Crampton Lake have mean temperatures that are statistically distinct from all other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: A two-sample t-test looking at if the mean temperature of Peter Lake is significantly different than the mean temperature of Paul Lake.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```

#filter dataset
Crampton_Ward_LakeTTest <-
  NTL_LTER_subset %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))
#run ttest
t.test(data = Crampton_Ward_LakeTTest, temperature_C ~ lakename)

##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 5.0132, df = 1066.3, p-value = 6.264e-07
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## 1.073548 2.454401
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                14.19206                12.42808

```

Answer: The t-test returns that p-value is 6.264e-07 which is less than 0.05, thus the null hypothesis is rejected. The alternative hypothesis is accepted that the mean temperatures between Crampton Lake and Ward Lake are statistically different. This does match my answer for part 16, where Crampton Lake is one of two lakes that is statistically distinct from all other lakes.